

Practical Week-10

Task-1: Investigate inode structures using commands such as `ls -li`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

```
arikel@DESKTOP-AMUSJME:~/week5prac$ mkdir inode_task
arikel@DESKTOP-AMUSJME:~/week5prac$ pwd
/home/arikel/week5prac
arikel@DESKTOP-AMUSJME:~/week5prac$ echo "Hello inode"> original.txt
arikel@DESKTOP-AMUSJME:~/week5prac$ cat original.txt
Hello inode
arikel@DESKTOP-AMUSJME:~/week5prac$ ls -li original.txt
39231 original.txt
arikel@DESKTOP-AMUSJME:~/week5prac$ stat original.txt
  File: original.txt
  size: 12          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 39231         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ arikel)   Gid: ( 1000/ arikel)
Access: 2026-09-19 03:22:01.281403318 +0000
Modify: 2026-09-19 03:21:52.073404488 +0000
Change: 2026-09-19 03:21:52.073404488 +0000
 Birth: 2026-09-19 03:21:52.069404488 +0000
arikel@DESKTOP-AMUSJME:~/week5prac$ ln original.txt hardlink.txt
arikel@DESKTOP-AMUSJME:~/week5prac$ ls -li original.txt hardlink.txt
39231 hardlink.txt 39231 original.txt
arikel@DESKTOP-AMUSJME:~/week5prac$ stat original.txt
  File: original.txt
  size: 12          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 39231         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ arikel)   Gid: ( 1000/ arikel)
Access: 2026-09-19 03:22:01.281403318 +0000
Modify: 2026-09-19 03:21:52.073404488 +0000
Change: 2026-09-19 03:24:01.785373786 +0000
 Birth: 2026-09-19 03:21:52.069404488 +0000
```

```
Change: 2026-09-19 03:24:01.785373786 +0000
Birth: 2026-09-19 03:21:52.069404488 +0000
arikel@DESKTOP-AMUSJME:~/week5prac$ echo "Modified through hard
link" >> hardlink.txt
arikel@DESKTOP-AMUSJME:~/week5prac$ cat original.txt
Hello inode
Modified through hard link
arikel@DESKTOP-AMUSJME:~/week5prac$ ln -s original.txt symlink.
txt
arikel@DESKTOP-AMUSJME:~/week5prac$ ls -li original.txt hardlin
k.txt symlink.txt
39231 -rw-r--r-- 2 arikela arikela 39 Sep 19 03:26 hardlink.txt
39231 -rw-r--r-- 2 arikela arikela 39 Sep 19 03:26 original.txt
39232 lrwxrwxrwx 1 arikela arikela 12 Sep 19 03:28 symlink.txt -
> original.txt
arikel@DESKTOP-AMUSJME:~/week5prac$ stat symlink.txt
  File: 'symlink.txt' -> 'original.txt'
  size: 12             Blocks: 0          IO Block: 4096   symb
olic link
Device: 8,48    Inode: 39232      Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ arikela)   Gid: ( 1000/
arikela)
Access: 2026-09-19 03:28:34.713289384 +0000
Modify: 2026-09-19 03:28:08.677298761 +0000
Change: 2026-09-19 03:28:08.677298761 +0000
Birth: 2026-09-19 03:28:08.677298761 +0000
arikel@DESKTOP-AMUSJME:~/week5prac$ find . -type f
./producer_consumer
./producer_consumer.c
./task_test.txt
./ls_grep.c
./original.txt
./ls_grep
./hardlink.txt
arikel@DESKTOP-AMUSJME:~/week5prac$ find . -type l
./symlink.txt
arikel@DESKTOP-AMUSJME:~/week5prac$ ls -li
```

```

arikela@DESKTOP-AMUSJME:~/week5prac$ ls -li
total 52
39231 -rw-r--r-- 2 arikela arikela    39 Sep 19 03:26 hardlink.t
xt
39216 drwxr-xr-x 2 arikela arikela  4096 Sep 19 03:20 inode_task
39194 -rwxr-xr-x 1 arikela arikela 16304 Sep 12 06:24 ls_grep
39195 -rw-r--r-- 1 arikela arikela  1346 Sep 12 06:23 ls_grep.c
39231 -rw-r--r-- 2 arikela arikela    39 Sep 19 03:26 original.t
xt
39192 -rwxr-xr-x 1 arikela arikela 16360 Sep 12 05:31 producer_c
onsumer
39193 -rw-r--r-- 1 arikela arikela  1645 Sep 12 05:30 producer_c
onsumer.c
39232 lrwxrwxrwx 1 arikela arikela    12 Sep 19 03:28 symlink.tx
t -> original.txt
39196 -rw-r--r-- 1 arikela arikela     0 Sep 12 06:02 task_test.
txt
arikela@DESKTOP-AMUSJME:~/week5prac$ |

```

Task-2: Design a program using mmap() to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and Implementation complexity. lets complete these tasks too

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GNU nano 8.7.1mmap_file.c *

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <string.h>

#define FILE_NAME "mmap_data.txt"
#define SIZE 100

int main()
{
    int fd;
    char *data;

    // Open/create file
    fd = open(FILE_NAME, O_RDWR | O_CREAT, 0666);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    // Set file size
    if (ftruncate(fd, SIZE) == -1)
    {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    // Map file into memory
```

^G Help

^X Exit

^O Write Out

^R Read File

^F Where Is

^_ Replace

^K Cut

^U Paste

^T Execute

^J Justify

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GNU nano 8.7.1

mmap_file.c *

```
// Map file into memory
data = mmap(NULL, SIZE,
             PROT_READ | PROT_WRITE,
             MAP_SHARED,
             fd, 0);

if (data == MAP_FAILED)
{
    perror("mmap");
    close(fd);
    return 1;
}

printf("File mapped into memory.\n");

// Write through memory mapping
strcpy(data, "Hello from mmap!");

printf("Data written using mmap: %s\n", data);

// Make changes visible to the file
if (msync(data, SIZE, MS_SYNC) == -1)
{
    perror("msync");
}

// Unmap memory
munmap(data, SIZE);

close(fd);

| printf("Memory unmapped and file closed.\n");
```

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^ Replace

^K Cut
^U Paste

^T Execute
^J Justify

```

// Unmap memory
munmap(data, SIZE);

close(fd);

printf("Memory unmapped and file closed.\n");

return 0;
}
|

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

```

arikela@DESKTOP-AMUSJME:~/week5prac$ nano mmap_file.txt
arikela@DESKTOP-AMUSJME:~/week5prac$ nano mmap_file.c
arikela@DESKTOP-AMUSJME:~/week5prac$ gcc mmap_file.c -o mmap_file
arikela@DESKTOP-AMUSJME:~/week5prac$ ./mmap_file
File mapped into memory.
Data written using mmap: Hello from mmap!
Memory unmapped and file closed.
arikela@DESKTOP-AMUSJME:~/week5prac$ cat mmap_data.txt
Hello from mmap!arikela@DESKTOP-AMUSJME:~/week5prac$

```